



INDRA KUMAR

Applied AI Engineer | Generative AI | RAG | LLM Fine-Tuning

Indian Institute of Science, Bengaluru

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Professional Summary

Applied AI Engineer with an M.Tech in Robotics and Autonomous Systems from IISc, currently working in Product Development & R&D at OLA. Experienced in Retrieval-Augmented Generation, LLM fine-tuning, parameter-efficient adaptation, predictive maintenance, and production ML pipelines using Python, PyTorch, Hugging Face Transformers, FastAPI, Docker, and Azure.

Technical Skills

Programming & Data	Python, SQL, C++, Pandas, NumPy, scikit-learn, XGBoost
GenAI & LLMs	Retrieval-Augmented Generation (RAG), LoRA, QLoRA, PEFT, Hugging Face Transformers, embeddings, semantic search, vector search, hybrid retrieval
ML & Deep Learning	PyTorch, TensorFlow, classification, regression, anomaly detection, CNNs, Vision Transformers, U-Net, reinforcement learning
MLOps & Cloud	FastAPI, REST APIs, MLflow, Docker, Git, GitHub Actions, Azure AI Search, Azure Blob Storage, Azure Container Apps
Optimization & Robotics	Gurobi, Google OR-Tools, A* path planning, PID control, autonomous navigation

Professional Experience

OLA **Jun 2026 – Present**
Product Development and R&D *Bengaluru, India*

- Built an end-to-end RAG pipeline for industrial-equipment manuals using PyMuPDF, Azure Blob Storage, and Azure AI Search, covering PDF extraction, semantic chunking, embeddings, and incremental indexing.
- Implemented keyword, dense-vector, and hybrid retrieval with semantic reranking; evaluated Recall@5, MRR, grounded-answer accuracy, and citation accuracy, and fine-tuned an instruction-following model using QLoRA/PEFT with page-level citations, confidence-based abstention, and hallucination guardrails.
- Developed containerized FastAPI services for Azure Container Apps and contributed to Project Safai, modernizing internal platforms and reducing Salesforce dependency.

Transportation Networks Laboratory, IISc **Sept 2024 – May 2026**
Graduate Student Researcher – M.Tech Thesis and EV Research Projects *Bengaluru, India*

- Under **Prof. Tarun Rambha**, developed a two-stage stochastic mixed-integer Gurobi model jointly optimizing charger retrofits, charging schedules, request acceptance, and dynamic parking/energy pricing under demand uncertainty; evaluated 180- and 363-scenario datasets from ACN-Caltech sessions, California electricity prices, and charger-cost data, identifying profit-optimal conversion levels of 5–14% for a 100-space facility.
- Under **Prof. Prasant Misra** (TCS Distinguished Senior Scientist, TCS Research), formulated EV routing with last-mile delivery and grid-energy discharge under time-window, battery, payload, station, distance, and fleet constraints; developed constraint-aware Maskable PPO and Linear Q-learning and benchmarked extended 100-customer Solomon instances across cost, fleet usage, runtime, grid discharge, and price sensitivity.

Math Expert, Photomath **Jan 2022 – May 2024**
Subject Matter Expert *Remote / Part-time*

- Produced rigorous step-by-step LaTeX solutions in Linear Algebra, Calculus, Probability, and Optimization, and reviewed peer submissions for mathematical correctness, rigor, pedagogy, formatting, and adherence to quality rubrics.

Selected Projects

Multimodal Brain Tumor Segmentation with LoRA-Adapted Transformers [\[Project Link\]](#) **Feb 2025 – May 2025**

- Built 2D/3D BraTS 2021 segmentation pipelines for WT, TC, and ET from FLAIR, T1, T1CE, and T2 MRI; adapted Swin-UNETR and ViT-Small using LoRA on QKV/projection layers with frozen transformer weights and Dice/BCE loss.
- Compared slice-based, tumor-centered 3D patch, and full-volume strategies with five-fold cross-validation; achieved 0.857 mean Dice with patches and 0.901 WT Dice with resizing using only 0.67–0.82M trainable parameters in the 2D variants.

Path Planning and Control for Wheeled Robots Using Aerial Imagery [\[Project Link\]](#) **Sep 2025 – Oct 2025**

- Built a TurtleBot4 navigation pipeline combining TensorFlow/Keras U-Net road segmentation, grid-based A* planning, PID control, and ROS; generated binary traversability maps using a four-level encoder-decoder with skip connections, batch normalization, and 0.1 dropout.
- Extended A* to eight-neighbor search with diagonal costs and octile heuristic for smoother paths; tuned linear/angular PID controllers using cross-track and heading errors and demonstrated navigation on a physical TurtleBot4.

Intelligent Automated OT Scheduler – NPCI Hackathon Winner **Jan 2026**

- Built an autonomous operating-theatre scheduling system for hospital resource allocation, surgeon scheduling, equipment constraints, emergency cases, cancellations, and real-time rescheduling.

- Used Google OR-Tools CP-SAT to solve NP-hard scheduling constraints and developed an XGBoost-based surgery duration prediction module for adaptive planning.
- Achieved 95% OT utilization through optimized scheduling and dynamic rescheduling during operational disruptions.

AMD Hackathon – Reasoning Q&A Agents

Jul 2025

- Fine-tuned Qwen-3-4B using LoRA on a 25K reasoning dataset to build specialized LLM-based question and answer agents for multi-step reasoning.
- Applied GRPO-based optimization and Hugging Face Transformers workflows to improve reasoning quality over supervised fine-tuning baselines.

GPT-2 From Scratch – LLM Systems [\[Project Link\]](#)

Jun 2025

- Reimplemented the 124M-parameter GPT-2 decoder in PyTorch with masked multi-head attention, residual connections, layer normalization, and tied embeddings; loaded the official checkpoint and verified numerical agreement with Hugging Face logits.
- Built autoregressive generation and domain fine-tuning pipelines using cross-entropy loss, AdamW, learning-rate scheduling, and mixed-precision training.

Building-Type Classification from Google Street View [\[Project Link\]](#)

Aug 2025 – Dec 2025

- Trained VGG16 transfer-learning models on 2,516 grayscale street-view images across five structural-material classes for earthquake-risk assessment, with resizing, RGB conversion, normalization, and spatial augmentation.
- Compared baseline VGG16, Canny-edge preprocessing, and embedding-similarity filtering; diagnosed overfitting, class imbalance, and majority-class bias using learning curves, confusion matrices, and Kaggle scores.

Modeling Strategic Bidding Styles in Online Auctions [\[Project Link\]](#)

Jan 2026 – Apr 2026

- Engineered six behavioral features from 6,321 bidder-auction records across 807 auctions; derived incrementalist, jump-bidder, and sniper labels using K-means and trained multinomial logit, achieving 99.5% accuracy and 0.995 weighted F1.
- Modeled bidder heterogeneity using mixed logit with 2,000 simulation draws and three-component Gaussian mixtures; trained a PyTorch DQN for 5,000 simulated auctions, attaining 65% and 91% win rates against mixed opponents and jump bidders, respectively.

Understanding A3C – Convergence and Linear Speedup in Reinforcement Learning

Feb 2025 – Apr 2025

- Studied A3C convergence under general function approximation and softmax policies, including the $\mathcal{O}(\epsilon^{-2.5}/N)$ per-worker sample-complexity bound and linear speedup with N asynchronous workers.
- Validated convergence and sample-efficiency behavior experimentally on synthetic environments, OpenAI Gym tasks, and Atari games.

Education

M.Tech in Robotics and Autonomous Systems <i>Indian Institute of Science, Bengaluru</i>	2024 - 2026
B.Tech in Electrical Engineering <i>University of Lucknow</i>	2017 - 2021

Relevant Coursework

Machine Learning for Signal Processing | Reinforcement Learning | Deep Learning for NLP | Deep Learning for Computer Vision | Data Structures and Algorithms | Computational Methods of Optimization | Linear Algebra | Mathematical Techniques for Robotics | Foundations of Robotics

Achievements

- Secured **All India Rank 451** in GATE 2024 Electrical Engineering among 59,599 candidates.
- Won **Gold/First Place in Chess** at the IISc Intra-Departmental Championship and represented the department on board one during the finals.